

A Structural Break, Not a Weaker Cohort

Explaining Albania's 2022 PISA Collapse with
Survey-Weighted Machine Learning

PISA 2009–2022 Data Science Project

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Abstract

Background. Albania’s share of 15-year-olds below PISA mathematics Level 2 rose from 42% in 2018 to 75% in 2022, the sharpest single-cycle regression in the assessment. We ask whether machine learning on background-questionnaire data can identify at-risk students, why they are at risk, and whether the 2022 collapse is merely a lower-scoring cohort or a change in the risk structure itself.

Methods. Using all five Albanian PISA cycles (2009–2022; 27,042 students) and eight comparison countries (323,121 students), we apply the survey methodology PISA requires: final student weights on every estimate, plausible values combined by Rubin’s rules, and balanced repeated replication for design-based standard errors. We compare survey-weighted classifiers, add school socioeconomic composition features, cross-check a multilevel model and a direct school-questionnaire linkage, and audit calibration, fairness, coverage and a difference-in-differences.

Results. A domain classifier separates 2018 from 2022 students with AUC 0.98 on background features alone, so the collapse is a structural covariate shift, not just lower scores. Individual background variables hit a ceiling (cross-validated AUC 0.73) that tuning does not move; adding school socioeconomic composition lifts every model to ~ 0.78 and makes gradient boosters beat logistic regression, exposing a *feature* ceiling. The new ~ 0.78 plateau is then a genuine *data* ceiling: a multilevel model (school ICC ≈ 0.31) and a school-questionnaire linkage (no gain) converge on it, it recurs across all nine systems with its size tracking each system’s between-school ICC ($r \approx 0.80$), and richer within-test process data reaches ~ 0.86 only through a signal endogenous to the score. The crisis is broad-based: Albania has the flattest socioeconomic gradient of the nine countries, shares its risk drivers with peers, and a difference-in-differences with a failed placebo shows the collapse is national, not local to the 2019 earthquake.

Conclusions. The 2022 collapse is a structural break in the risk-generating process, not a weaker cohort. We release a calibrated, interpretable screener, framed as associational; a naive threshold over-flags the poorest quintile (false-positive-rate gap 0.63), which group-specific thresholds close to 0.003, and partial-identification bounds under 79% coverage keep the crisis far above 2018.

Keywords: PISA; educational data mining; low mathematics proficiency; covariate shift; survey-weighted machine learning; algorithmic fairness; conformal prediction; Albania

July 2026

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1. Introduction

In four years, three in four Albanian fifteen-year-olds crossed the line into low mathematics proficiency. The OECD Programme for International Student Assessment (PISA) measures whether students can apply reading, mathematics, and science to real problems, and its *Level 2* boundary marks basic proficiency; students below it are the international “low performers” that education systems target. In PISA 2018, 42% of Albanian students fell below the mathematics Level 2 line. In PISA 2022, 75% did. No other participating system regressed so far in a single cycle. A drop of this magnitude invites an obvious reading: the 2022 cohort was simply weaker, and the country needs broad remediation. Whether that reading is right is not a semantic question. It determines whether policy should treat 2022 as more of the same, only worse, or as evidence that *who* is at risk, and *why*, has changed.

This paper treats the question as one of educational data mining, and it insists on the survey methodology that PISA requires. PISA is a stratified, clustered sample reported through final student weights, and each achievement score arrives as ten *plausible values*, multiple imputations of a latent proficiency. Analyses that ignore the weights do not estimate a population quantity, and analyses that collapse the plausible values to a single number understate their uncertainty. A large body of secondary PISA work nonetheless does both. Every estimate below is survey-weighted; plausible values are combined by Rubin’s rules; and where the data permit (the 2015, 2018, and 2022 cycles carry 80 balanced-repeated-replication weights) standard errors are

design-based.

From the student background questionnaire, socioeconomic status, home possessions, parental education and occupation, grade repetition, immigration status, school belonging, teacher support, math anxiety, and access to information technology, we ask three questions. Can a classifier identify Albanian students at risk of low mathematics proficiency, and how well? Which factors drive that risk, and do they differ from the drivers in comparable countries? And does the 2022 collapse represent a change in the risk *structure* that can explain the 2018-to-2022 reversal? The economic and structural conditions of Albania in this exact window, a destructive 2019 earthquake, an extended pandemic school closure, a thin remote-learning infrastructure, and sustained emigration of working-age adults and teachers, give these questions a concrete backdrop that we return to in Section 6.

Thesis. Albania’s 2022 mathematics collapse is not a uniformly weaker cohort but a *structural break in the risk-generating process*, driven by a disruption to the supply of schooling rather than to household resources; the students at risk are identifiable from background data only up to a genuine ceiling of about 0.78 AUC that three independent methods confirm, their risk is broad-based rather than concentrated in the poor, and its drivers are the same ones that operate across the region.

Roadmap. Section 3 describes the five-cycle data and the design-based estimation. Section 4 sets out the leakage-safe, survey-weighted modelling and the covariate-shift diagnostic. Section 5 reports the four empirical findings: the structural shift (AUC 0.98), the feature-versus-data ceiling, the shared and broad-based driver structure, and the calibrated, fairness-audited screener. Section 6 then reads those findings against Albania’s economic and structural conditions, arguing that a supply-side shock, not a household-income shock, best fits the evidence. Section 7 discusses implications and limits, and Section 8 returns to the collapse with which we began.

Contributions. Our central contribution is methodological and, we believe, transferable beyond this dataset: we show that the accuracy ceiling recurring across PISA prediction studies is not one phenomenon but two, a movable *feature* ceiling and a fixed *data* ceiling, and that the two can be told apart with convergent evidence rather than with another round of tuning. Applied to Albania, this lets us (i) show the 2022 collapse is a structural covariate shift rather

than a level change; (ii) locate the background-data accuracy ceiling, demonstrate it is a feature ceiling by moving it with school socioeconomic composition, certify the new plateau as a genuine data ceiling through three independent routes, and show it both generalises across nine systems (its size tracking each system’s between-school inequality) and withstands richer process data that only moves it through a signal endogenous to the test; (iii) characterise the crisis as broad-based and shared with regional peers, and test a single-region earthquake explanation with a difference-in-differences that instead identifies the collapse as national; and (iv) connect these statistical findings to Albania’s economic and structural context, while delivering a calibrated, interpretable, coverage-bounded and fairness-mitigable risk screener that we argue should not be read causally.

2. Background and Related Work

PISA survey methodology. PISA reports each student’s proficiency as ten plausible values drawn from a posterior latent-trait distribution, and every published statistic is weighted by the final student weight, with sampling variance estimated by balanced repeated replication (BRR) (OECD, 2023b, 2009). Rubin’s rules combine the point estimates and the between-imputation variance across the plausible values (Rubin, 1987). These are not optional refinements: using a single averaged score as a target understates uncertainty, and unweighted statistics do not estimate a population quantity. A large share of secondary PISA analysis nonetheless drops both conventions, which, as we argue below, is part of why the accuracy figures it reports should be read with care. We follow the manual throughout.

Machine learning on PISA background data. Predicting achievement or at-risk status from PISA background questionnaires is by now a well-populated literature, and it converges on a consistent picture in which tree ensembles dominate. Masci, Johnes and Agasisti (2018) analyse PISA 2015 across nine countries with a two-stage tree-based model and recover socioeconomic status, grade repetition and school climate as leading correlates, while stressing that much of the variation sits at the school rather than the student level. On the most recent cycle, Djekourmane et al. (2026) predict mathematics scores across six East Asian systems in PISA 2022 and find gradient boosting strongest: XGBoost reaches $R^2 \approx 0.57$ and a thresholded AUC near 0.81, with LightGBM close behind, and, tellingly, the linear baselines trail badly on R^2 yet come within a few points on AUC. Erdoğan and Taştan (2024) recover the same ordering

on the Turkish PISA 2018 subset, where a boosted regression tree edges out ridge, LASSO, bagging and random forests. For the post-pandemic window most relevant to us, Delprato (2025) models low-performing students across ten Latin American systems in PISA 2022 and uses SHAP attributions to rank their determinants, foregrounding interpretation over any single accuracy number. Throughout, XGBoost, LightGBM and CatBoost (Chen and Guestrin, 2016; Ke et al., 2017; Prokhorenkova et al., 2018) are the recurring strong baselines on this tabular regime, and SHAP values (Lundberg and Lee, 2017) the usual explanation layer.

Two regularities in this work frame our own contribution. First, reported performance sits in a narrow band, roughly AUC 0.78 to 0.84, or accuracy in the high seventies, largely irrespective of country, cycle, or the exact ensemble chosen. Second, the strong learners and the linear baselines are usually separated by a fraction of an AUC point, yet that gap is presented as a ranking rather than tested for significance, and even studies that expose it directly (linear models trailing on R^2 while nearly tying on AUC; Djekourmane et al., 2026) stop short of asking whether the difference is real or what would move it. The literature reports where the accuracy lands but rarely diagnoses why it lands there.

Accuracy ceilings, and what they are made of. That recurring ceiling near AUC 0.8 is usually treated as a fixed backdrop, a number to beat with a better learner or more tuning. We instead make it the object of study, and this is the paper’s central methodological contribution. A plateau in predictive accuracy can have two very different causes. It may be a *feature ceiling*, in which a model has exhausted the signal in the feature families it was given but a richer family (here, the socioeconomic composition of a student’s school) would move it; or it may be a *data ceiling*, in which the measurement instrument itself, the background questionnaire, cannot resolve risk any further and no feature engineering or model class will help. The two are indistinguishable from a single learning curve, and the reflex of tuning harder or adding capacity cannot separate them, because it holds the feature family fixed. We argue the distinction is nonetheless diagnosable, but only with *convergent evidence*: a feature ceiling is demonstrated when a new feature family lifts every model together, and a data ceiling is demonstrated when independent routes (a hand-crafted composition model, a design-appropriate multilevel model, and a direct school-questionnaire linkage) all plateau at the same value. We are not aware of a prior PISA study that draws this line explicitly, which is why the field’s shared ~ 0.8 has gone largely uninterrogated.

Within-country geographic disparities. Cross-national PISA work reports one figure per country, and the country-level literature on Albania documents an urban-rural achievement gap in broadly qualitative terms (World Bank, 2014): rural students trail urban ones, and the gap is attributed to socioeconomic development. What is missing, for Albania and for its Balkan peers, is a *design-based* decomposition of low-proficiency risk *within* the country. The public-use file carries no subnational region variable, which is likely why such a decomposition has not been reported; the geographic signal is instead encoded in the explicit sampling stratum, which labels every school by an urbanicity, region-band and sector cell. To our knowledge, this paper is the first to recover those cells and report the survey-weighted below-Level-2 rate for each, with design-based standard errors and a significance test on every between-group gap, turning a qualitatively known disparity into a quantified, targetable map. The multilevel literature above establishes that much PISA risk variance is between schools (Masci et al., 2018; Goldstein, 2011); we locate *which* schools, geographically.

Dataset shift. When the input distribution changes between two populations, a classifier trained to tell them apart quantifies the change, and a high domain-classifier AUC is direct evidence of covariate shift (Quiñonero-Candela et al., 2009). We use this as the central diagnostic for the 2018-to-2022 reversal rather than relying only on marginal score distributions, which cannot separate a merely lower-scoring cohort from one whose risk structure has moved.

Calibration, decision curves, and fairness. A screening probability must be calibrated to be actionable. Isotonic regression is a standard non-parametric post-hoc calibrator (Zadrozny and Elkan, 2002), and decision-curve analysis evaluates a screener by net benefit across operating thresholds rather than by accuracy at a single cut (Vickers and Elkin, 2006), which matters when prevalence is high. Group-fairness diagnostics such as demographic parity and equalized odds expose disparate error rates across protected groups (Hardt et al., 2016). We report all three, survey-weighted, a combination seldom applied together in the PISA setting.

Nested design and honest comparison. PISA students are nested in schools, so a random-intercept (multilevel) logistic model is the design-appropriate specification, and its intraclass correlation quantifies the between-school variance (Goldstein, 2011). Finally, because repeated cross-validation folds share training data, naive fold-score intervals are anti-conservative; we

use the Nadeau–Bengio corrected-resampled variance (Nadeau and Bengio, 2003) and compare models with the corrected resampled t -test (in-sample) and the DeLong test on shared test predictions (DeLong et al., 1988). This is the machinery that turns the field’s informal ranking of near-tied models into a tested claim about which gaps are real.

3. Data

3.1 Datasets

We use the student questionnaire files for all five PISA cycles from 2009 to 2022. The 2015/2018/2022 cycles ship as SPSS `.sav`; 2009 and 2012 are fixed-width ASCII with SPSS control files, which we parse with a purpose-built reader. After harmonising feature names across cycles, the Albanian longitudinal file holds **27,042** students. A comparison set of nine countries (Albania, North Macedonia, Montenegro, Serbia and Bulgaria as Balkan peers; Estonia and Finland as top performers; and Mexico and Colombia as GDP-matched economies) holds **323,121** students. Raw files are multi-gigabyte and not redistributed; processed slices are regenerated by the pipeline.

3.2 Target variable

A student is *at risk* if their mathematics score falls below the exact OECD Level 2 boundary of 420.07 points. Because each score arrives as ten plausible values $PV_1, \dots, PV_{10}$, the model-comparison target is the point (PV-mean) indicator,

$$AT_RISK = \begin{cases} 1 & \text{if } \frac{1}{10} \sum_{j=1}^{10} PV_j < 420.07 \quad (\text{below Level 2}) \\ 0 & \text{otherwise (proficient),} \end{cases}$$

while the final estimates use per-plausible-value targets combined by Rubin’s rules. In the 2022 crisis cohort the class balance is **75.3% at risk** versus **24.7% proficient**.

3.3 Features

The core set is thirteen background variables with high cross-cycle availability: ESCS (the OECD socioeconomic index), home possessions, parental education (HISCED) and occupation

Table 1: Analytic data dictionary. OECD indices are standardised (mean 0, SD 1 across OECD). “Cycles” lists the Albanian cycles with non-missing values.

Variable	Meaning	Direction	Cycles
MATH_PV_MEAN	Math score, mean of 10 plausible values	higher = better	all 5
AT_RISK	Low proficiency: math < 420 (below Level 2)	1 = at risk	all 5
ESCS	Economic, social & cultural status index	higher = advantaged	'09,'18,'22
HOMEPOS	Home possessions (material + educational)	higher = more	all but '15
HISCED	Highest parental education (ISCED level)	higher = more	'09,'18,'22
HISEI	Highest parental occupation status (ISEI)	higher = higher	'09,'18,'22
BELONG	Sense of belonging at school	higher = more	all but '15
TEACHSUP	Perceived teacher support in maths	higher = more	'12,'18,'22
ANXMAT	Mathematics anxiety	higher = more anxious	'12,'22
REPEAT	Has repeated a grade	1 = repeated	all but '15
IMMIG	Immigrant background	1 = 1st/2nd gen	all but '15
GRADE	Grade relative to modal grade	0 = modal	all 5
GENDER	Student gender	coded 0/1	all 5

(HISEI), gender, grade repetition, immigration status, school belonging, teacher support in mathematics, math anxiety, grade relative to modal, and home/school ICT resources. Composite and interaction features (a standardised SES composite, a material-deficit rank, digital-readiness indices, and centred interaction terms) are derived *inside* the cross-validation pipeline on the training fold only, so their scaling and rank statistics never leak from the test fold.

3.4 Data dictionary and provenance

Table 1 names each analytic variable, its plain-English meaning, the direction of its scale, and the cycles in which Albania actually carries it. The OECD background indices (ESCS, HOMEPOS, BELONG, ...) are standardised to an OECD mean 0 / SD 1, so a value of -0.75 reads directly as “0.75 SD below the OECD student average” before any modelling.

3.5 Survey-weighted descriptive summary

Table 2 is the population-level “know your data” view of the 2022 crisis cohort: weighted mean and SD, the raw range, the weighted median, and how much is missing. Two facts stand out. First, Albania sits far below the OECD SES origin (ESCS -0.75 , HOMEPOS -0.87). Second, the attitudinal indices carry heavy missingness (34–40% for teacher support and math

Table 2: Survey-weighted descriptive summary, Albania 2022. Mean/SD/median use final student weights; min/max are raw sample extremes.

Variable	<i>n</i>	%miss	w-mean	w-SD	median	range
MATH_PV_MEAN	6129	0.0	368.2	79.0	357.5	142–693
ESCS	5575	9.0	−0.75	1.09	−0.77	−5.2–2.8
HOMEPOS	5699	7.0	−0.87	1.02	−0.89	−6.3–5.4
HISCED	4603	24.9	5.60	2.51	5.0	1–10
HISEI	4773	22.1	45.2	23.5	31.7	11–89
BELONG	4743	22.6	0.25	1.05	0.14	−3.3–2.8
TEACHSUP	4002	34.7	0.41	1.25	0.82	−2.9–1.6
ANXMAT	3674	40.1	0.01	1.17	0.05	−2.4–2.6
GRADE	4734	22.8	−0.04	0.35	0.0	−3–2

Table 3: Weighted effect sizes vs. the at-risk target, Albania 2022, ranked by magnitude. Numeric: Cohen’s *d*; categorical: Cramér’s *V*.

Feature	Measure	Value	<i>p</i>
HOMEPOS	Cohen’s <i>d</i>	−0.504	n/a
ESCS	Cohen’s <i>d</i>	−0.496	n/a
ANXMAT	Cohen’s <i>d</i>	+0.486	n/a
HISEI	Cohen’s <i>d</i>	−0.455	n/a
BELONG	Cohen’s <i>d</i>	−0.231	n/a
TEACHSUP	Cohen’s <i>d</i>	−0.141	n/a
REPEAT	Cramér’s <i>V</i>	0.104	< 0.001
GENDER	Cramér’s <i>V</i>	0.074	< 0.001
GRADE	Cohen’s <i>d</i>	−0.044	n/a
HISCED	Cohen’s <i>d</i>	−0.040	n/a
IMMIG	Cramér’s <i>V</i>	0.027	0.161

anxiety), which caps how much any model leaning on them can contribute, a caveat that recurs downstream.

3.6 Which features separate at-risk students?

Ranking predictors by the *strength* of their weighted association with low proficiency, not just by significance (since at $n \approx 6,000$ almost everything is “significant”), gives Table 3. Numeric features use weighted Cohen’s *d* (at-risk vs. proficient; sign shows direction), categoricals use Cramér’s *V*. Home possessions, ESCS and *math anxiety* are the strongest separators ($|d| \approx 0.5$, medium); the demographic categoricals are trivially small ($V < 0.1$), and immigrant status is not even significant ($p \approx 0.16$). Risk here is an SES-and-affect story far more than a demographic one, and it sets the bar a later feature-importance analysis (§5) must beat.

Table 4: Sample composition and comparability, 2022 (survey-weighted), sorted by at-risk rate. Colombia and Mexico are poorer yet less at-risk than Albania.

Country	N	mean score	mean ESCS	at-risk %	ESCS %miss
Albania	6129	368.2	−0.75	75.3	9.0
Colombia	7804	382.7	−1.07	72.6	4.4
Mexico	6288	395.0	−0.95	67.2	0.3
North Macedonia	6610	388.6	−0.28	67.1	3.0
Montenegro	5793	405.6	−0.21	60.2	1.5
Bulgaria	6107	417.3	−0.27	54.0	4.0
Serbia	6413	439.9	−0.20	43.2	1.0
Finland	10239	484.1	+0.26	24.2	3.0
Estonia	6392	509.9	+0.15	13.8	1.6

3.7 Cross-country comparability

Before ranking countries we ask whether the 2022 samples are comparable (Table 4). They are broadly matched in size and ESCS coverage, so the ranking that follows is not an artefact of one country being thinly or selectively measured. The table also states the paper’s puzzle numerically: Colombia and Mexico sit at a *lower* mean ESCS than Albania yet post *lower* low-proficiency rates. SES level alone therefore does not explain Albania’s position; the gradient’s slope, examined in §5, does the work.

3.8 The socioeconomic gradient as a probability curve

Figure 1 shows the continuous gradient, $P(\text{low proficiency} \mid \text{ESCS})$, as a descriptive weighted-logistic curve with the weighted at-risk rate per SES decile overlaid, for 2018 versus 2022. The 2022 curve sits far above 2018 at *every* SES level: the crisis raised risk across the board, not only for the disadvantaged. The gradient (slope) persists, so SES still matters, but a pure “poverty shock” story would predict a steepening the data only partly show, consistent with the feature-drift result in §5. This is a single-predictor descriptive gradient, distinct from the multivariable models below.

3.9 Design-based estimates

The 2015/2018/2022 cycles carry 80 Fay balanced-repeated-replication weights; we combine the BRR sampling variance with the between-plausible-value variance to report the standard error the OECD manual prescribes. The 2009/2012 fixed-width cycles do not carry replicate weights and fall back to an imputation-only standard error, which we flag. Table 5 gives Albania’s

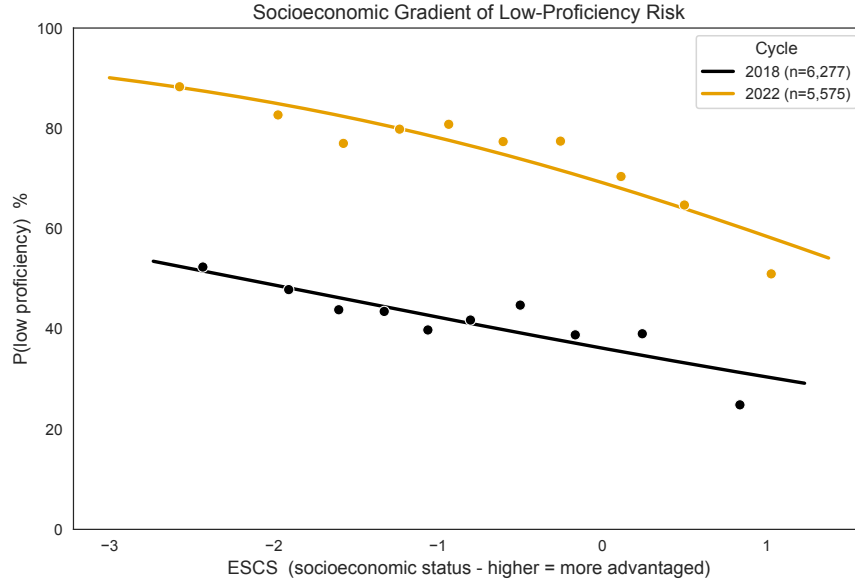


Figure 1: Descriptive socioeconomic gradient of low-proficiency risk, 2018 vs. 2022. Markers are weighted at-risk rates per ESCS decile; lines are weighted logistic fits. The 2022 curve is shifted up at every SES level.

Table 5: Albania weighted at-risk mathematics rate (below Level 2) with design-based standard errors. BRR available only for the SPSS cycles.

Cycle	At-risk	SE	95% CI	BRR
2009	0.677	0.0076	0.663–0.692	n/a
2012	0.607	0.0051	0.597–0.617	n/a
2015	0.533	0.0108	0.512–0.554	✓
2018	0.424	0.0088	0.407–0.441	✓
2022	0.739	0.0041	0.731–0.747	✓

207 weighted at-risk trajectory with these design-based standard errors: a V-shape that falls from
 208 68% (2009) to 42% (2018) and then snaps back to 74% in 2022 (Figure 2).

209 3.10 A verified data gap

210 Albania genuinely lacks the ESCS index in 2012 and 2015 (an OECD coverage gap confirmed
 211 against the source files, not a processing bug), whereas all peer countries have full 2015 back-
 212 ground data. This limits Albania’s *own* cross-cycle socioeconomic comparison, but it leaves the
 213 2018-versus-2022 shift analysis, in which both cycles are fully covered, unaffected.

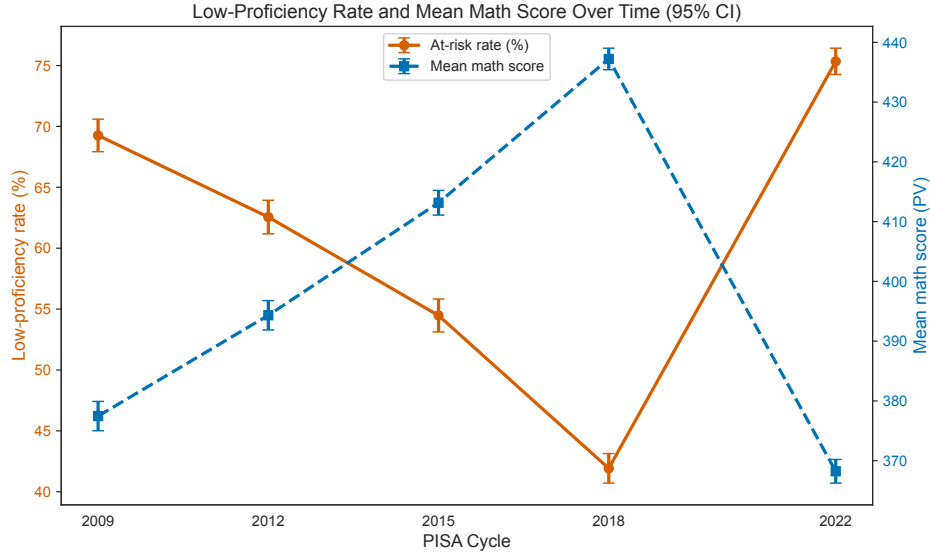


Figure 2: Albania’s weighted below-Level-2 mathematics rate, 2009–2022. The 2018 low (42%) and the 2022 spike (74%) bracket the structural break this paper explains.

4. Methods

4.1 Survey-correct estimation

Every descriptive statistic and every model metric is weighted by the final student weight W_{FSTUWT} . Where a model is fit, weights are routed to the estimator’s own `sample_weight` (correcting a subtle bug in which scaler-wrapped models, logistic regression, SVM, KNN, MLP, silently fell back to an unweighted fit because the weight was passed to the wrong pipeline step). Class-conditional statistics rescale the population-scale weights to the effective sample size before, e.g., a χ^2 test, so significance is not inflated by the weight magnitudes.

4.2 Leakage-safe pipeline

Each fold’s pipeline is: (1) an engineered-feature builder that learns component means, standard deviations, and the home-possession rank curve on the training fold and replays them on the test fold; (2) median imputation fit on the training fold; (3) the estimator. Interaction terms are built from mean-centred components so a product is not confounded with its main effects. School-composition features are survey-weighted, *leave-one-out* school means (a student’s own value is excluded from their school mean), and a fold-safe re-computation that recomputes the means from the training fold only reproduces an identical effect, so the lift is compositional, not leakage.

4.3 Evaluation and comparison

Models are scored with repeated stratified 5×4 cross-validation, weighted ROC and PR AUC, F1-macro, and MCC. Fold-score confidence intervals use the Nadeau–Bengio corrected-resampled variance, which replaces the naive $1/k$ term with $1/k + n_{\text{test}}/n_{\text{train}}$ to account for the correlation between folds that share training data. Model-vs-model differences are tested directly: the corrected resampled t -test on paired per-fold AUCs (cross-validation) and the DeLong test on shared test-set predictions (out-of-sample). This turns a ranking table into a claim about which gaps are real.

4.4 Covariate-shift diagnostic

To quantify the 2018→2022 change we train a domain classifier to distinguish the two cohorts from background features alone, with weighted subsampling, pooled-median imputation, and missingness indicators; a high AUC certifies that the input distribution, not just the score level, moved. We validate the diagnostic on a null (same cohort split at random → AUC ≈ 0.5).

4.5 Explanation, calibration, fairness, forecast

We compute TreeSHAP global and per-student attributions on the school-context model, and partial-dependence / centred-ICE curves for the top drivers. For deployment we add out-of-fold isotonic calibration and tune the decision threshold on calibrated out-of-fold predictions to maximise weighted MCC. The fairness audit reports weighted demographic parity, equal-opportunity (TPR) and false-positive-rate gaps across gender, SES quintile, and immigration status, plus a threshold sweep. A random-intercept logistic model provides the design-appropriate specification and the between-school intraclass correlation. Finally, a scenario Monte-Carlo forecast for 2026 propagates each cycle’s design-based standard error through a weighted trend fit; with only five cycles and a structural break we report scenarios and their intervals, not a point prediction.

4.6 Robustness and extension analyses

Five further analyses probe the main claims. (i) *Cross-country ceiling replication*: the student-only versus school-context comparison is re-run identically in all nine systems, and each system’s between-school ICC is read from its random-intercept model, to test whether the fea-

ture ceiling and its size generalise. (ii) *Modality ablation*: PISA’s computer-based cognitive-process indicators are added to the school-context model and split into cognitive-process and questionnaire-process blocks, so an endogenous within-test signal can be separated from data available at screening time. (iii) *Fairness mitigation*: group-specific decision thresholds are fit on calibrated out-of-fold scores to equalise the per-quintile false-positive rate (equalized-odds post-processing), tracing the accuracy–equity frontier. (iv) *Partial identification*: given the 79% coverage index, worst-case and monotone Manski bounds on the population at-risk rate are computed, the monotone bound assuming the out-of-frame group is no better off than the assessed. (v) *Difference-in-differences*: a design-based DiD contrasts the earthquake-affected central region with the rest of the country across 2018 and 2022, with a 2015→2018 placebo to test parallel trends; weights, plausible values, and replicate-weight standard errors are carried throughout.

4.7 Geographic equity and lever ranking

Two descriptive-to-prescriptive analyses target the crisis rather than just measure it. For *geographic equity*, the sampling stratum label is parsed into a region-band $\times$ urbanicity $\times$ sector cell for the 2015/2018/2022 cycles; the weighted below-Level-2 rate and its design-based (replicate-weight and plausible-value) standard error are computed per cell, and each between-group gap is estimated on the contrast itself so its standard error is exact under the design rather than inferred from two marginal intervals. For the *lever ranking*, the fitted school-context model’s features are partitioned into fixed endowments and malleable levers; each lever is nudged half a training-standard-deviation in the direction that lowers the smooth mean predicted probability (its engineered interactions and school-mean aggregate moving with it), and levers are ranked by the resulting change in the weighted population predicted at-risk rate. Each lever effect carries a population-bootstrap 95% confidence interval (400 resamples of the students with the fitted model held fixed, so the interval captures finite-cohort sampling uncertainty but not model-fit uncertainty). A fixed-versus-malleable bundle contrast sizes how far actionable levers reach before the structural floor. Like the policy simulations these are model-associational contrasts, not causal effects.

4.8 Reproducibility and engineering

Logic lives in unit-tested modules (189 tests on synthetic fixtures, covering the weighted statistics, Rubin’s rules, the leakage-safe transformer, the BRR+PV variance, the Manski bounds and group-threshold mitigation, and the Nadeau–Bengio / DeLong maths); notebooks are narrative layers over them. On macOS the boosters (Homebrew `libomp`) and scikit-learn (bundled OpenMP) can abort a process if their runtimes collide; we isolate each model fit in a fresh subprocess that imports the boosters before scikit-learn, which resolved an import-order segfault and returned LightGBM to the comparison.

5. Results

5.1 The 2022 collapse is a structural shift

A domain classifier separates 2018 from 2022 Albanian students with weighted AUC **0.98** on background features alone. The 2022 cohort is not simply a lower-scoring version of 2018 along a stable structure; the input distribution itself moved. The most-shifted variables (Figure 3) are teacher support in mathematics (standardised mean difference -0.31) and home possessions ($+0.27$), with school belonging, grade repetition, and ESCS following. That teacher support fell while reported home possessions rose is consistent with a disruption to the *schooling process* rather than a household-wealth shock, and it foreshadows the compositional result below. Out-of-sample the shift is detectable but not catastrophic: models trained on 2009–2018 still transfer to 2022 at $\text{AUC} \approx 0.67$ (covariate shift without total collapse).

5.2 An accuracy ceiling, and its cause

On the thirteen student features, twelve estimators cluster tightly (Figure 4). CatBoost leads at weighted CV AUC 0.731, but a paired Nadeau–Bengio test finds the top three statistically *tied*: CatBoost $\approx$ logistic regression ($p = 0.50$) $\approx$ gradient boosting ($p = 0.37$): a transparent linear model matches the best boosters. Nested-CV Optuna tuning of the top three yields no significant gain (CatBoost $+0.002$, $p = 0.81$; GBM $+0.004$, $p = 0.45$; LR -0.001 , $p = 0.43$). This looks like a data ceiling.

It is not; it is a *feature* ceiling. Adding survey-weighted, leave-one-out **school-mean** ESCS,

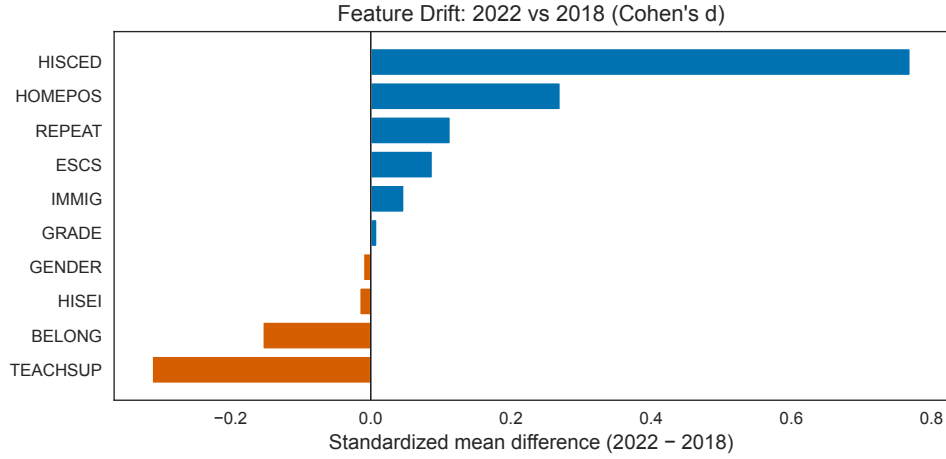


Figure 3: Standardised 2018→2022 drift of Albanian background features. Teacher support and home possessions move most; the domain classifier that separates the cohorts reaches AUC 0.98.

Table 6: Albania 2022, weighted 5×4 CV, *with* school context. Δ AUC is the lift over the student-only baseline.

Model	AUC	Δ AUC	PR-AUC	F1	MCC
CatBoost	0.783	+0.052	0.907	0.693	0.393
Gradient Boosting	0.777	+0.050	0.905	0.665	0.365
LightGBM	0.775	+0.060	0.902	0.688	0.381
Random Forest	0.769	+0.060	0.895	0.644	0.345
Extra Trees	0.758	+0.051	0.884	0.641	0.332
XGBoost	0.756	+0.058	0.892	0.657	0.333
Logistic Regression	0.754	+0.027	0.897	0.649	0.332
SVM (RBF)	0.749	+0.037	0.894	0.609	0.282
MLP (128-64)	0.743	+0.034	0.884	0.620	0.295
Naive Bayes	0.701	+0.019	0.856	0.610	0.263
KNN	0.700	+0.036	0.861	0.613	0.257
Decision Tree	0.609	+0.041	0.797	0.608	0.217

home possessions, anxiety, and teacher support, plus cohort size, lifts every model by ~ 0.05 AUC (Table 6): CatBoost $0.731 \rightarrow \mathbf{0.783}$, and the effect is identical under fold-safe re-computation. Crucially, adding school context *breaks the tie*: the boosters now beat logistic regression significantly (GBM vs LR $p = 0.001$), because they exploit a student $\times$ school-composition interaction a linear model cannot. SVM and MLP, added for completeness, land mid-table: the podium is a booster story, not an artefact of omitting kernel or neural learners.

5.3 ~ 0.78 is a genuine data ceiling

Three independent routes converge on ~ 0.78 . (i) The composition booster above plateaus there. (ii) A random-intercept logistic model, the design-appropriate specification, *matches* the hand-

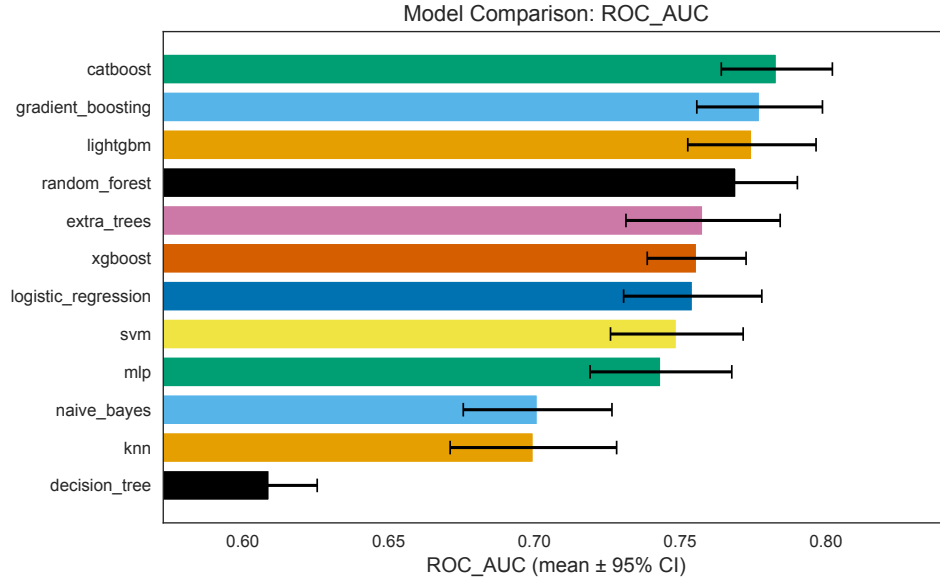


Figure 4: Weighted CV AUC with Nadeau–Bengio intervals. Boosters form a top band once school context is included; the interpretable logistic-regression baseline gains least because it cannot use the student $\times$ school interaction.

crafted school-mean booster on identical folds (CV AUC ≈ 0.78) and reports a school intraclass correlation ICC ≈ 0.31 : about a third of risk variance is *between* schools and barely touched by student features. (iii) Linking the PISA school questionnaire (principal-reported resources, staff, class size, leadership) directly onto students adds no significant signal beyond composition (best +0.004 AUC, all $p > 0.24$), because school-mean SES already proxies the school inputs. The ceiling is a property of what background questionnaires can measure, not of tuning or model capacity. An ensemble stack of the four boosters confirms this: it reaches AUC 0.786 vs single CatBoost 0.783 (+0.003, $p = 0.03$): statistically significant, practically nil, and it *regresses* MCC and F1 at $\sim 19\times$ the compute. Single CatBoost stays the deployable headline. The ceiling is not an Albanian idiosyncrasy. Replaying the identical student-only versus school-context comparison across all nine systems (Table 7) reproduces the pattern almost everywhere: the composition lift is significant in eight of nine countries (the exception is low-inequality Estonia, $p = 0.09$), and its *size* tracks how much risk variance sits between schools. The correlation between a system’s school intraclass correlation and its school-context AUC is $r \approx 0.80$; where schools are socioeconomically segregated (Bulgaria ICC 0.56, North Macedonia 0.40) composition buys the most, and where they are equitable (Finland ICC 0.10) it buys almost nothing (+0.008 AUC). The feature-versus-data ceiling is thus a general property of background-questionnaire prediction, and the amount a system can move it is set by its own

Table 7: Student-only vs. school-context AUC across the nine 2022 systems. The composition lift is significant everywhere but Estonia and scales with the between-school ICC ($r \approx 0.80$ between ICC and school-context AUC).

Country	Group	Base	School	Lift	ICC
Albania (ALB)	Balkan	0.714	0.773	+0.059	0.31
North Macedonia (MKD)	Balkan	0.769	0.835	+0.066	0.40
Montenegro (MNE)	Balkan	0.734	0.817	+0.083	0.34
Serbia (SRB)	Balkan	0.745	0.805	+0.060	0.36
Bulgaria (BGR)	Balkan	0.791	0.857	+0.066	0.56
Colombia (COL)	GDP-matched	0.834	0.863	+0.028	0.45
Mexico (MEX)	GDP-matched	0.749	0.794	+0.045	0.31
Estonia (EST)	Top performer	0.729	0.752	+0.022	0.21
Finland (FIN)	Top performer	0.781	0.790	+0.008	0.10

between-school structure rather than by the learner.

5.4 The ceiling is a measurement limit, not a modelling one

If the ~ 0.78 plateau is a property of what the *background* questionnaire can see, then a genuinely richer modality should move it, whereas more of the same questionnaire should not. Both hold. Adding PISA’s computer-based cognitive-process data (the 15 timing-, action-, and effort-derived indicators from the assessment delivery itself) lifts the school-context model from AUC 0.773 to **0.858** ($+0.085$, $p < 0.001$). But almost all of that gain is the cognitive-process block: cognitive features alone reproduce it ($+0.078$), while the non-cognitive *questionnaire*-process features add only $+0.011$ ($p = 0.02$). The cognitive-process signal is endogenous to the outcome, it is measured *during* the same test whose result we predict, so it inflates apparent accuracy without being available at screening time. The deployable ceiling, over data a school could hold *before* a student sits the assessment, therefore stays at ~ 0.78 . Richer data can move the plateau, but only data that is not a downstream trace of the score itself, which confirms the ~ 0.78 figure as a measurement limit rather than a modelling one.

5.5 Drivers: anxiety and school composition, shared with peers

On the school-context model, four of the top seven SHAP drivers are school-level (Figure 5): school-mean home possessions leads, followed by individual math anxiety, school-mean teacher support, and school-mean ESCS. The model reads a student’s risk more from the socioeconomic *composition* of their school than from their own resources; this is the compositional effect made

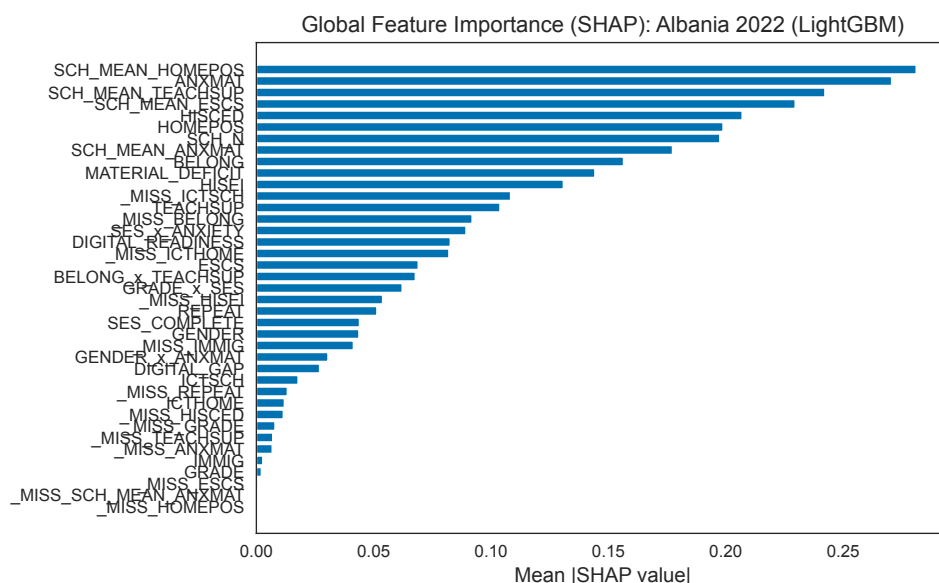


Figure 5: Global SHAP importance (school-context model). Four of the top seven drivers are school-level; math anxiety is the strongest individual factor.

explicit. Immigration status is negligible. A feature $\times$ country SHAP rank matrix (Figure 6) shows these drivers are *shared*: math anxiety is a top-3 driver almost everywhere (including Estonia), and school composition dominates most systems. What is distinctive about Albania is the *level and breadth* of low proficiency, not the mechanism.

5.6 A broad-based, flat-gradient crisis

Per-country school-context models (Figure 7) place Albania at the highest at-risk rate (75%) but one of the *lowest* model AUCs (0.77): with three in four students at risk there is little contrast to separate. Albania also has the **flattest** SES gradient of the nine countries (~ 17 score points per SD), a floor effect: even advantaged Albanian students score near 369, close to the line. Estonia and Finland have *steeper* gradients ($\sim 38\text{--}40$) yet far higher levels, because there is more score range for SES to predict. The policy reading is direct: a flat gradient plus school-composition dominance means SES-targeted transfers alone miss most at-risk Albanian students; system-wide levers (teacher support, math-anxiety reduction, raising the floor across all schools) are required.

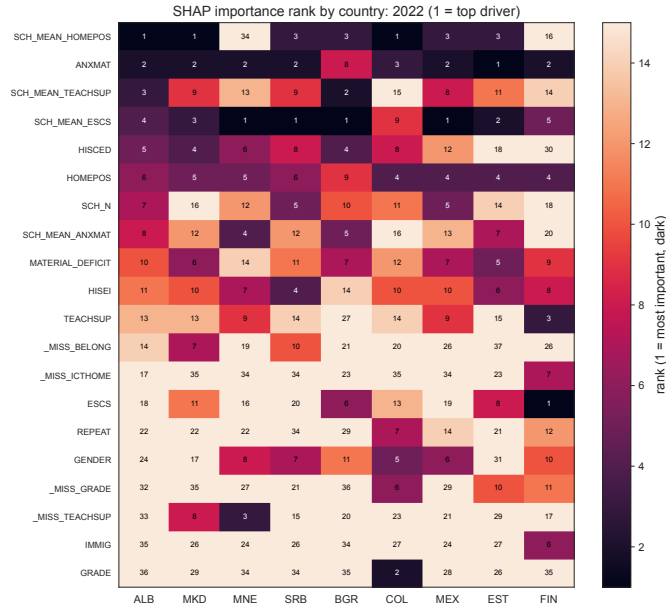


Figure 6: SHAP driver ranks by country. Math anxiety and school composition recur across systems; Albania mirrors its peers in mechanism.

5.7 The geography of the crisis: where *inside* Albania it concentrates

A flat SES gradient tells a policymaker that family income is a weak targeting axis; it does not say what a better one is. Albania’s public-use file carries no subnational region, but the sampling stratum labels every school by a geographic cell, *region band* (North / Center / South) $\times$ *urbanicity* (Urban / Rural) $\times$ *sector* (Public / Private), which we decode to map the below-Level-2 rate *within* the country across 2015, 2018 and 2022. Every rate is survey-weighted with a design-based standard error (replicate weights and plausible values by Rubin’s rules), and each between-group *gap* is estimated on the contrast itself so it carries a significance test rather than being read off two overlapping marginal intervals.

The load is sharply uneven (Figure 8). The North band is the highest in every cycle, and the rural cells carry the heaviest burden: in 2022 rural North (0.83) and rural Center (0.82) are the worst cells, while even urban Center, which contains the capital Tirana, is the *lowest* cell at 0.69, still a majority at risk. The gaps are real under design-based inference: in 2022 the rural-minus-urban gap is +9.9 points ($p \approx 10^{-12}$), North minus South is +7.0 ($p \approx 3 \times 10^{-10}$), and public minus private is +25.5 ($p < 0.001$), the widest of the three and *growing* (0.15 in 2018 to 0.26 in 2022), though private schools are a small, self-selected $\sim 13\%$ of students, so that gap is a composition signal rather than a like-for-like school effect. The 2018→2022 reversal was broad-based, every cell rose ~ 28 – 37 points, but it stacked on top of pre-existing rural disadvantage,

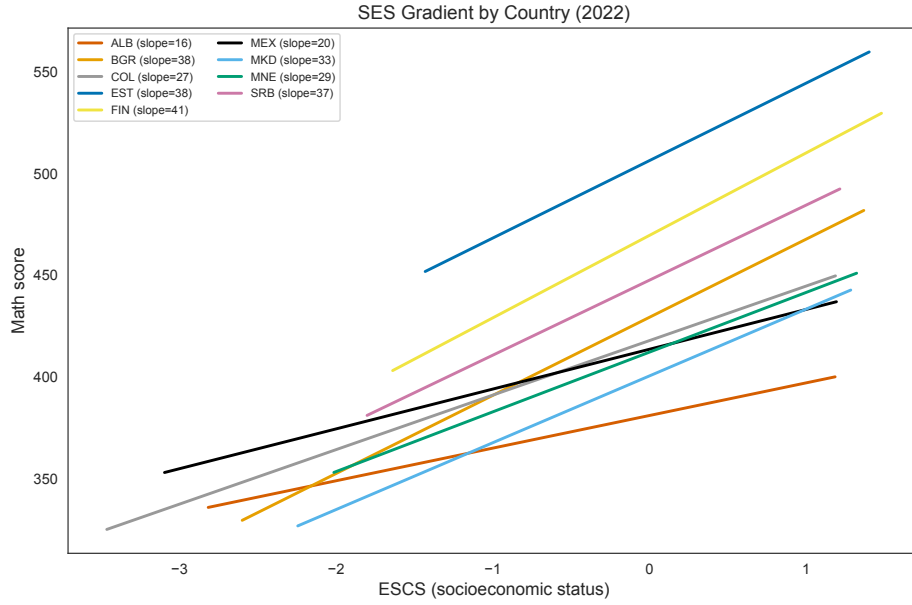


Figure 7: SES gradient of mathematics score, 2022. Albania’s slope is the flattest of the nine countries, a floor effect, not a steep advantage gap.

so the rural bands ended highest. Because the SES gradient is flat, income-targeted transfers alone miss most at-risk students; **geography is a sharper targeting axis than family SES**, and the North, rural and public-school bands are where a fixed remediation budget reaches the most at-risk students.

5.8 From prediction to prescription: which levers can policy move?

Global SHAP ranks features by how much the model *reads* them, but it mixes two kinds. *Fixed endowments*, family SES, parental education, home possessions, immigrant status, gender and grade, describe *who* is at risk but move under no short-run policy; *malleable levers*, math anxiety, teacher support, belonging, grade-repetition exposure and ICT access, are the things a school system can actually change. We split the two and rank only the actionable levers by the reduction in the population predicted at-risk rate from a standardised half-SD nudge in the beneficial direction, with the direction discovered on the smooth mean predicted probability rather than the noisy thresholded count, and each lever’s effect carries a population-bootstrap 95% confidence interval (400 resamples of the students, the fitted model held fixed). Among the levers, **math anxiety is the one with real reach** (Figure 9): a half-SD reduction cuts the predicted at-risk rate by 5.1 points (95% CI [4.5, 5.7]) and teacher support by 1.0 ([0.6, 1.5]), the only two intervals that exclude zero; belonging, ICT and grade-repetition are essentially flat. The interval reflects sampling uncertainty in the finite cohort and, by design, not model-fit

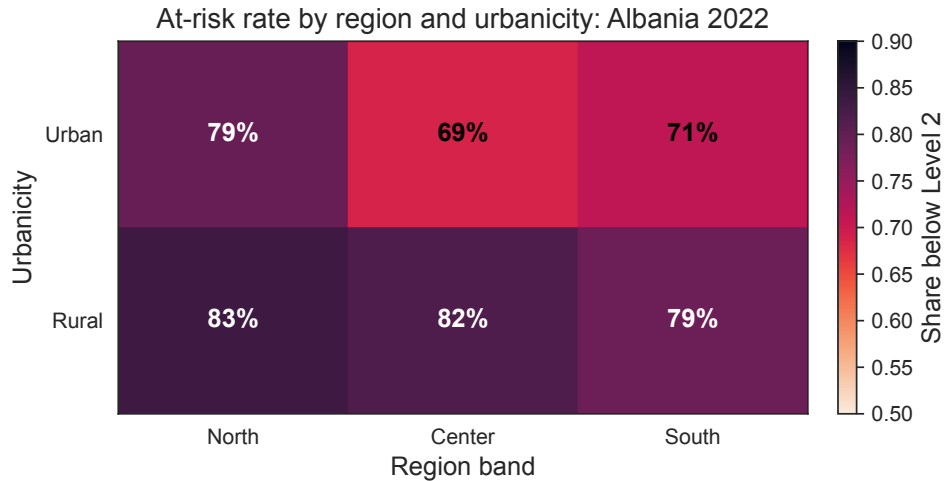


Figure 8: At-risk rate by region band and urbanicity, Albania 2022 (survey-weighted; darker is higher). Rural North and rural Center carry the heaviest load; urban Center, which contains Tirana, is the lowest cell yet still a majority at risk.

uncertainty, which the frozen model does not propagate.

The exercise also makes the structural floor explicit. Moving *every* actionable lever half an SD at once takes the predicted at-risk rate from ~ 0.72 to ~ 0.66 (-6.9 points), and a hypothetical that lifts every fixed endowment by the same amount, which no short-run policy can deliver, buys only ~ 5 points more. Both bundles leave a large majority at risk: the crisis is structural, not a tuning problem, the same conclusion the covariate shift, the flat gradient and the school-composition result reach from other directions. Levers and geography are complements: the anxiety lever is *weaker where the crisis is deepest*, its reduction is far smaller in the rural band than the urban one, so anxiety-reduction programmes help most in urban schools while the rural and northern floor needs system-level investment. As with the policy simulations, these are model-associational contrasts, not causal effects, and are read as such.

5.9 A calibrated, fair-audited screener

Raw CatBoost probabilities are sharp but mis-calibrated (weighted ECE ≈ 0.12); out-of-fold isotonic calibration cuts this to ≈ 0.01 (Figure 10), and decision-curve analysis shows the model beats screen-everyone / screen-no-one across the selective threshold band, the relevant test at 75% prevalence. The fairness audit is less comfortable (Figure 11): at a fixed threshold the screener over-flags the poorest SES quintile (false-positive rate ≈ 0.86 vs ≈ 0.24 for the richest; parity gap 0.48, FPR gap 0.63). Because school-mean SES *is* SES, making it a top feature doubly encodes the gradient; adding school context shrinks the gender gap but leaves the SES

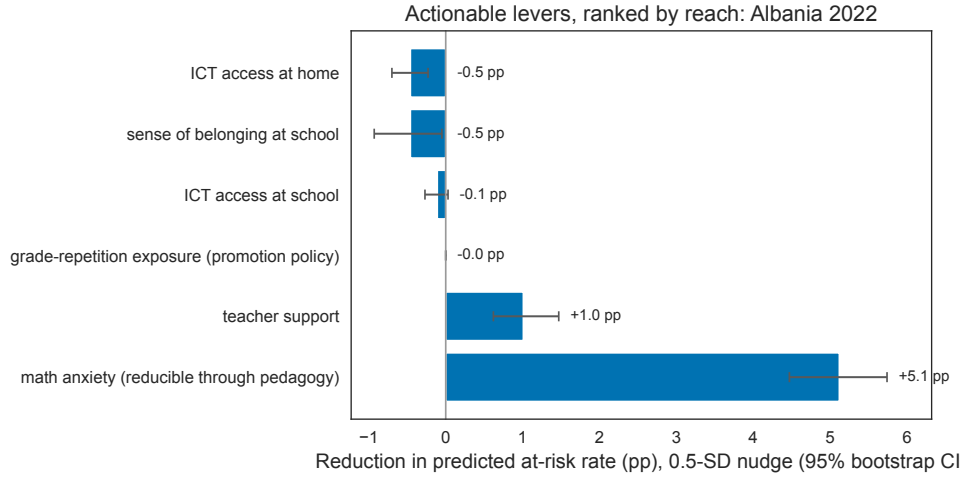


Figure 9: Actionable levers ranked by the reduction in the weighted predicted at-risk rate from a 0.5-SD nudge in the beneficial direction (Albania 2022). Math anxiety dominates; the remaining levers are near-flat.

gap intact. This is an equalized-odds concern.

The gap is, however, fixable at the operating point. Post-processing the calibrated scores with *group-specific* thresholds (a per-quintile cut chosen to equalise the false-positive rate, the equalized-odds remedy) collapses the SES FPR gap from 0.63 to **0.003** while holding the overall false-positive rate fixed. It is not free: matching the poorest quintile’s threshold to the rest trades recall (overall 0.81 $\rightarrow$ 0.75) and accuracy (0.75 $\rightarrow$ 0.70) for equity, the familiar fairness–accuracy tension made explicit rather than hidden. We therefore ship the single-threshold screener as the default but expose the group-threshold frontier, and frame either as a tool to *widen* support, never to ration it.

5.10 Which students can the screener call? Conformal prediction sets

A calibrated probability answers *how likely*, but not *how sure*: at 75% prevalence a middling score is a weak signal dressed as a precise number. Split (inductive) conformal prediction turns the calibrated CatBoost score into a *prediction set* carrying a finite-sample coverage guarantee, $P(y \in \hat{C}(x)) \geq 1 - \alpha$, that holds under exchangeability *without* assuming the model is correct. On a stratified 60/20/20 proper-train / calibration / test split we score each calibration student by the non-conformity $1 - \hat{p}(y | x)$ of their true label, take the survey-weighted $[(n+1)(1-\alpha)]/n$ quantile of those scores, and admit into a test student’s set every label whose probability clears the induced floor; the weights make the guarantee hold for the *population*, not the unweighted sample. The read is operational: a *singleton* set ($\{\text{at risk}\}$ or $\{\text{not at risk}\}$) is a confident call,

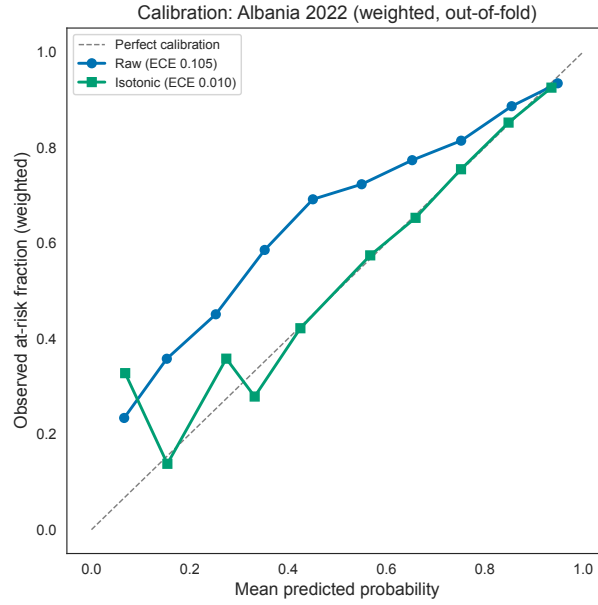


Figure 10: Reliability curve before/after out-of-fold isotonic calibration; weighted ECE falls from ≈ 0.12 to ≈ 0.01 .

and the *both-labels* set is the model abstaining.

At the 90% coverage level ($\alpha = 0.1$) the guarantee is met on held-out students (weighted coverage **0.925**, $n = 1226$) and the cohort splits cleanly: **60.0%** of students receive a confident singleton call and **40.0%** land in the both-labels abstention set, with no empty sets. The trade between confidence and coverage behaves as it should: tightening to 95% coverage drops the confident share to 44.3% (coverage 0.950), while loosening to 80% raises it to 88.4% (coverage 0.836). Class-conditional (Mondrian) calibration, which restores per-class coverage under the 75/25 imbalance, holds coverage in both classes but at the price of more abstention (singleton share 47.9% at the 90% level). Beyond the point screener this draws an explicit *competence boundary*: at a 90% guarantee the tool confidently labels three in five students and, instead of bluffing on the rest, hands the remaining two in five, the students nearest the decision boundary, to human judgement, which is precisely where an automated low-proficiency flag is least defensible. The screener therefore reports not only a calibrated risk but an honest account of which students it should not be trusted to call alone.

5.11 Is the 75% an artefact of coverage? Partial-identification bounds

Albania met only part of PISA's sampling standards in 2022, and its Coverage Index 3 is 0.79: roughly a fifth of the 15-year-old population is out of frame, mostly adolescents who have left or

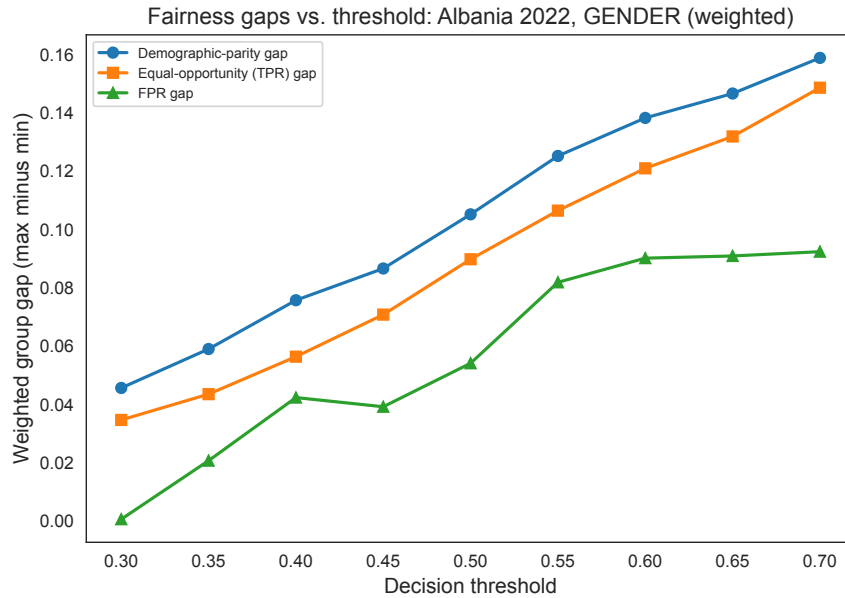


Figure 11: Weighted fairness gaps by decision threshold (gender shown). Gaps shift with the operating point rather than vanishing, mitigation needs constraints, not a knob.

never entered school. Because that missing fifth is plausibly *worse* off, the headline rate could in principle be an underestimate, so we ask what the data alone can guarantee without assuming anything about the uncovered. The assessed sample’s weighted at-risk rate is 0.739 (design-based 95% CI 0.731–0.747). Worst-case Manski bounds, letting every uncovered student take any value, give [0.584, 0.794]; the more defensible monotone bound, assuming the out-of-frame group is at least as at-risk as those assessed, gives [**0.739**, **0.794**]. Under either identifying assumption the 2026-relevant crisis level stays far above Albania’s 2018 rate of 0.42: coverage counsels caution on the exact percentage but cannot explain the collapse away, and if anything the true breadth is wider than the measured ~ 0.75 .

5.12 Is the collapse local to the earthquake? A difference-in-differences

The 2019 Durrës–Tirana earthquake is the one shock Albania did not share with its peers, so it is tempting to credit the collapse to it. A design-based difference-in-differences tests this directly, comparing the earthquake-affected central region against the rest of the country across the 2018 and 2022 cycles (final weights, plausible values by Rubin’s rules, 80 replicate weights). The affected region’s at-risk rate did rise 4.7 points more than the control’s ($\text{DiD} = +0.047$, $p < 0.001$), which in isolation looks like a local earthquake effect. It is not identified: a *placebo* DiD on the pre-earthquake 2015→2018 window is itself large and significant with the opposite sign (-0.070 , $p < 0.001$), so the two regions were already on diverging trajectories

and the parallel-trends assumption fails. The credible reading is the one the magnitudes give directly, that both regions collapsed by roughly the same enormous amount (treated +0.34, control +0.30): the crisis is *national*, not concentrated at the epicentre. This is a well-identified *null* for a local causal story and independent corroboration of the broad-based, system-wide account, the pandemic-era disruption to schooling reached every region, and no single-region natural experiment can carry the explanation.

5.13 Forecast for 2026

With only five cycles and a 2022 structural break, a point forecast is indefensible, so we report a Monte-Carlo scenario band with design-based standard errors propagated. The honest message is its width: the defensible zone for the 2026 below-Level-2 rate runs from persistence of the crisis (median 0.74, 90% PI 0.66–0.82) through partial reversion (0.47, 0.43–0.52) to a full pre-COVID recovery (0.21, 0.18–0.24), and even the optimistic tail sits above Albania’s 2018 low of 42%. We treat the forecast as context for the structural findings above, not as a headline result.

6. The Economic and Structural Context of the Collapse

The statistical results locate the collapse in time and shape but not in cause; the economic and structural history of Albania between the 2018 and 2022 assessments supplies the missing narrative. PISA 2018 was fielded in early 2018 and PISA 2022 in 2022, so any explanation must point to forces that acted in that four-year window. Two shocks did, and only one of them was shared with Albania’s peers.

Two compounding shocks hit the schooling window. Albania entered the pandemic already weakened by a natural disaster that its comparison countries did not face. On 26 November 2019 a magnitude 6.4 earthquake struck the Durrës and Tirana region, the strongest to hit the country in more than four decades; it killed 51 people, injured about 3,000, and damaged or destroyed housing and schools across roughly a dozen cities in the most populous part of Albania (Government of Albania and European Union and United Nations and World Bank, 2020). Barely three months later, COVID-19 closed schools nationwide from March 2020, and Albania recorded one of the longer cumulative closure durations in the region before in-

struction fully resumed (UNESCO, 2022; World Bank and UNESCO and UNICEF and USAID and FCDO and Bill & Melinda Gates Foundation, 2022). The comparison economies in this study lived through the pandemic, but none of them absorbed a major earthquake in the same window. Notably, the OECD’s own country note attributes Albania’s 2022 decline to exactly this pairing, an end-of-2019 earthquake that damaged schools and houses across twelve cities, compounded by the pandemic (OECD, 2023a). Albania therefore experienced a *double* disruption to the continuity of schooling precisely between the two assessments, which is where a structural break would have to originate.

The shock was to schooling, not to household wealth, and the data show it. If the 2022 collapse were a household-income shock, the socioeconomic markers in the questionnaire would have fallen; instead, the school-side marker fell while the household marker held. The covariate-shift analysis (Section 5) found that the variable which moved most from 2018 to 2022 was teacher support in mathematics, down by roughly a third of a standard deviation, while reported home possessions actually rose. A demand-side poverty shock would depress home possessions; the observed pattern instead points to a supply-side degradation of instruction, the signature of closed and damaged schools and of teaching capacity stretched thin. This interpretation is consistent with the macroeconomic record: Albania’s economy contracted in 2020 and rebounded strongly the next year, with World Bank figures placing the contraction at roughly three to four percent and 2021 growth near nine percent as tourism recovered (World Bank, 2024a), while household consumption was cushioned throughout by diaspora remittances worth about nine percent of GDP (World Bank, 2024b). Family assets did not crater, which is why home possessions did not; the damage concentrated in the public delivery of schooling.

Structural weaknesses turned a temporary shock into a lasting shift. A closure need not become a permanent loss, but three structural features of Albania’s system made it likely to. First, remote learning depended on infrastructure that many households lacked: home internet and device access, especially in rural areas, left a substantial share of students unable to follow online instruction, so televised and online lessons reached pupils unevenly (World Bank and UNESCO and UNICEF and USAID and FCDO and Bill & Melinda Gates Foundation, 2022; UNICEF Albania, 2021). Second, Albania spends a comparatively low share of GDP on education, on the order of three to four percent, below the world average of about 4.6 percent

and below European Union norms (World Bank and UNESCO Institute for Statistics, 2024), leaving little slack to mount an intensive catch-up response. Third, Albania has for years experienced heavy emigration of working-age and skilled adults, including teachers; the 2023 census confirmed that the resident population had fallen by roughly 14 percent, about 420,000 people, since 2011, to some 2.4 million (INSTAT (Institute of Statistics of Albania), 2024). This brain drain thins instructional capacity at precisely the moment recovery demands more of it. Together these conditions explain why the disruption still registered, undiminished, in the 2022 scores rather than bouncing back.

The economic reading matches the statistical shape of the crisis. A system-wide capacity shock predicts a broad-based decline, and that is what the gradient analysis found. Because closures, damaged buildings, and thinned teaching capacity act on all students in a school rather than only its poorest, such a shock should depress even advantaged students and flatten the socioeconomic gradient. Albania in 2022 has the flattest SES gradient of the nine countries and a floor so low that even its advantaged students score near the proficiency line (Section 5); its mean mathematics score fell from 437 in 2018 to 368 in 2022, a drop of about 69 points (OECD, 2023a). A shock concentrated in household poverty would have steepened the gradient, not flattened it. A system-wide shock also predicts a *geographically* broad decline, and a difference-in-differences confirms it: the mathematics collapse is no larger in the earthquake-affected central region than elsewhere once a placebo exposes diverging pre-trends (Section 5), so the 2019 earthquake compounded but did not localise the loss, the decisive disruption was the national, pandemic-era interruption of schooling. The convergence of the economic history and the statistical shape is the central insight of this section: the collapse behaves like a supply-side disruption to schooling, not a demand-side collapse in family means.

A caveat on measured effort. One qualification tempers the magnitude, though not the direction, of the drop. The OECD country note reports that Albanian students spent less effort on the 2022 assessment than in 2018 and shows signs of lower test engagement (OECD, 2023a), so part of the measured score fall may reflect test-taking behaviour rather than learning alone. This does not undo the structural reading. Our central shift diagnostic (Section 5) is trained on background questionnaire responses, not on the scores themselves, and still separates the 2018 and 2022 cohorts at AUC 0.98; the change in *who* the students are is real independent of how

hard they tried on test day.

A caveat on sample coverage. A second qualification concerns who the 2022 sample represents. The OECD flags that one or more of its sampling standards were not met for Albania in PISA 2022, and Albania’s Coverage Index 3, the share of the national 15-year-old population that the assessed sample represents, is about 79%: the 6,129 students tested in 274 schools stand in for roughly 28,400 of an estimated 36,000 15-year-olds (OECD, 2023b,a). The out-of-frame fifth is disproportionately made up of adolescents who have left school or were never enrolled, who as a group would be expected to perform below those still in the system. If anything, then, this exclusion biases the measured at-risk rate *downward*, so the true breadth of low proficiency is likely wider than the already severe 75% we report, and the collapse is not an artefact of sweeping in weaker students. We make this precise with partial-identification bounds (Section 5): even the worst-case Manski interval that makes no assumption about the uncovered fifth is [0.58, 0.79], and the monotone bound that treats them as no better off than the assessed is [0.74, 0.79], so under any defensible assumption the crisis rate stays far above Albania’s 2018 level of 42%. Two design points guard the central comparison further. Coverage was of comparable order in 2018, so the 2018-to-2022 contrast is not driven by a coverage change; and our structural-shift diagnostic conditions on background questionnaire responses within the assessed population rather than on population totals, so it is insensitive to the level of coverage even if it shifted. Coverage does, however, counsel caution in reading the absolute rate as a precise population figure, which is one more reason we lean on the shift in risk structure rather than on the headline percentage alone.

So what this implies for policy. Reading the collapse as broad-based and supply-side changes the policy prescription. If the problem were concentrated household poverty, socioeconomically targeted transfers would reach most at-risk students; because it is a flat, system-wide degradation of schooling capacity, such transfers alone would miss them. The levers the evidence points to instead are those that rebuild the delivery of instruction: retaining and paying teachers to slow the brain drain, completing physical reconstruction of damaged schools, funding sustained catch-up teaching, and addressing the two individual factors the model ranks highest, math anxiety and the quality of teacher support. These are the same system-wide levers that the shared, broad-based driver structure in Section 5 already implied, now grounded in the

country’s economic history.

7. Discussion

What the shift means. The single most important result is that the 2022 collapse is structural: a classifier tells 2018 from 2022 Albanian students almost perfectly from background data, and the variables that moved most are process variables: teacher support fell while reported home possessions rose. This is not the signature of a household-wealth shock; it is consistent with a disruption to schooling itself (the pandemic and its long tail), and it explains why models trained on the pre-crisis trajectory transfer only moderately to 2022. A “weaker cohort” framing would misdirect policy toward uniform remediation; a structural-shift framing points at the school process.

Why the ceiling matters. The instinct on hitting a plateau is to tune harder or add capacity. Here that instinct is wrong twice over. Tuning did nothing; capacity (a neural net, a kernel machine, a four-model stack) did nothing. The missing ingredient was a *feature family*, school socioeconomic composition, and once added, the plateau moves and the model class that wins changes. But the new plateau at ~ 0.78 is then real, corroborated by a multilevel ICC of ≈ 0.31 and by a direct school-questionnaire linkage that adds nothing. The lesson for educational data mining is methodological: distinguish a *feature* ceiling (movable, and diagnosable by what a richer feature family buys) from a *data* ceiling (a property of the measurement instrument), and use convergent evidence rather than a single learning curve to tell them apart. The distinction travels: the same feature ceiling appears in all nine systems and the amount each can move it scales with its between-school inequality ($r \approx 0.80$ with the ICC), and the data ceiling survives the obvious escape route, adding PISA’s within-test process data lifts AUC to ~ 0.86 but almost entirely through a cognitive-process signal endogenous to the score, leaving the deployable ceiling at ~ 0.78 . A plateau, then, can be moved by a new *feature family* or by a new *modality*, but only by data that a school could actually hold before the test.

Shared drivers, distinctive breadth. Albania’s risk mechanism is not exotic. Math anxiety and school composition drive risk across the nine countries; what distinguishes Albania is that low proficiency is *broad*, sitting on a flat SES gradient and a low floor. This reframes the policy conversation away from targeting the poor (whom a flat gradient says are not disproportionately

the whole problem) toward system-wide levers.

A single policy recommendation. The geography, lever and fairness results compose into one concrete, if associational, recommendation. *Target by place, not just by income.* Because the SES gradient is flat, means-tested transfers alone miss most at-risk students, whereas the North and rural bands carry a load 7 to 10 points above their comparators (and the public sector ~25 points above the small private one), so geography is where a fixed remediation budget reaches the most children. *Match the instrument to the place.* Math anxiety is the one soft lever with measurable reach (−5.1 points, 95% CI [4.5, 5.7]), but its reach is concentrated in urban schools and thins out in the rural band; so anxiety-oriented pedagogy (formative low-stakes assessment, teacher support) is a sensible *urban* first move, while the rural and northern floor, where no soft lever moves the needle and even lifting every endowment leaves a majority at risk, calls for structural investment (teaching capacity, learning time, school resources) rather than psychosocial programmes. *And screen to widen support, not to ration it.* Deploy the calibrated screener with the group-specific thresholds that equalise false-positive rates across SES, so the poorest quintile is not disproportionately flagged, and treat a flag as a route into these place-matched supports. The recommendation is deliberately singular: target the North, rural and public bands; lead with anxiety programmes in urban schools and structural investment in rural ones; and use an equity-corrected screener to direct the effort.

On causality, and the screener. Every model here is associational and fit on a single cross-section. The clearest illustration is anxiety: lower reported math anxiety co-occurs with higher scores, but it is partly a *consequence* of doing well, so a counterfactual that “reduces anxiety” in the model is not a defensible intervention estimate. We therefore ship the risk screener, calibrated probabilities, per-student SHAP drivers, and the fairness audit, as a *triage and communication* aid, not a lever simulator. The fairness result reinforces the point: a naive threshold over-flags the poorest quintile, so the tool is only defensible as a way to widen support; where equal false-positive treatment across SES is required, group-specific thresholds close the gap from 0.63 to 0.003 at a measured recall cost, an equity choice we surface rather than bury. The one causal question we could pose from observational data, whether the 2019 earthquake localised the collapse, we tested and answered in the negative: a difference-in-differences whose placebo fails identifies the loss as national, a well-identified null that guards against reading a

convenient single-region story into a country-wide shock.

Limitations. The student questionnaire is the only instrument here; we have no teacher or parent files, and the school questionnaire, while linked, adds no signal beyond composition. Albania lacks ESCS in 2012 and 2015, limiting its own cross-cycle socioeconomic comparison (the peer-benchmarked and 2018-vs-2022 analyses are unaffected). Design-based standard errors are available only for the 2015–2022 cycles; the fixed-width 2009/2012 cycles fall back to an imputation-only SE. The forecast rests on five cycles and one structural break, which is why we report scenarios rather than a number. Finally, the ~ 0.78 ceiling is specific to background-questionnaire prediction of a binary Level-2 target; we tested one richer modality, PISA’s within-test computer-based process data, and found it moves the plateau only through a signal endogenous to the score, so genuinely exogenous process data that a school holds before the assessment (attendance, formative assessment, longitudinal records) remains the untested route and the natural next step. The difference-in-differences likewise licenses only a negative causal claim; a design that could license a positive policy lever still requires data we do not have.

8. Conclusion

Three in four Albanian fifteen-year-olds were below basic mathematics proficiency in 2022, and the number with which we began now has an explanation. The collapse was not a uniformly weaker cohort but a change in the structure of who is at risk and why: a domain classifier separates the 2018 and 2022 cohorts almost perfectly from background data alone, the variable that moved most was the support students receive in school rather than the resources they have at home, and the decline sits on the flattest socioeconomic gradient of the nine countries we compare. Read against Albania’s history, a destructive 2019 earthquake, an extended pandemic closure, a thin remote-learning infrastructure, and the steady emigration of teachers and working-age adults, the statistical shape and the economic record tell the same story: a supply-side disruption to schooling, not a demand-side collapse in family means.

For anyone who would predict this risk, the practical lesson is where the ceiling comes from. Individual background answers plateau at about 0.73 AUC and no amount of tuning moves them; the single missing ingredient is the socioeconomic composition of a student’s school, which lifts every model to about 0.78, after which a multilevel model and a direct school-questionnaire

linkage independently confirm that the ceiling is now a property of what background questionnaires can measure rather than of the model. A plateau, in other words, can be a feature ceiling or a data ceiling, and telling them apart takes convergent evidence rather than one more round of optimisation.

For anyone who would act on it, the lesson is that a screener trained on a single cross-section reports associations, not levers. The same analysis that ranks math anxiety highly also shows why reducing a model input is not an intervention estimate, and the fairness audit shows a naive threshold over-flagging the poorest students. The tool we release is therefore built to widen support, never to ration it. The policy direction that follows is not targeted transfers to the poor, whom a flat gradient says are not the whole problem, but the rebuilding of schooling capacity for everyone: retaining teachers, reconstructing schools, and funding catch-up instruction.

Two questions that earlier framings left open we were able to take a first pass at here, and both answers point the same way. Richer data *can* move the 0.78 ceiling, PISA’s within-test process data reaches ~ 0.86 , but only through a signal endogenous to the score itself, so the deployable ceiling holds and the genuinely exogenous process records (attendance, formative assessment, longitudinal files) that would capture the disruption directly remain the route worth building. And the one causal question the observational data admit, whether the 2019 earthquake localised the collapse, returns a well-identified null: a difference-in-differences with a failing placebo shows the loss was national, not concentrated at the epicentre. What remains is a design that can license a *positive* policy lever rather than rule a single-region story out. Albania’s 2022 result is the clearest natural signal yet of what a compounded shock does to a fragile education system; understanding it well is the difference between planning for recovery and mistaking a broken process for a weak generation of students.

Reproducibility. Code, configuration, the unit-tested pipeline, the calibrated bilingual screener, and this manuscript’s figures are in the project repository. Raw PISA microdata are public from the OECD but not redistributed here.

List of abbreviations

ANXMAT	Mathematics anxiety index
AUC	Area under the receiver operating characteristic curve

BELONG	Sense of belonging (at school) index
BRR	Balanced repeated replication
CI	Confidence interval
CV	Cross-validation
DiD	Difference-in-differences
ECE	Expected calibration error
ESCS	(PISA index of) economic, social and cultural status
FPR	False-positive rate
GBM	Gradient boosting machine
GDP	Gross domestic product
HISCED	Highest parental education (in ISCED levels)
HISEI	Highest parental occupational status (ISEI)
HOMEPOS	Home possessions index
ICC	Intraclass correlation coefficient
ICE	Individual conditional expectation
ICT	Information and communication technology
IMMIG	Immigrant background
ISCED	International Standard Classification of Education
ISEI	International Socio-Economic Index of occupational status
KNN	k -nearest neighbours
MCC	Matthews correlation coefficient
MLP	Multilayer perceptron
OECD	Organisation for Economic Co-operation and Development
PI	Prediction interval
PISA	Programme for International Student Assessment
PR-AUC	Area under the precision–recall curve
PV	Plausible value
RBF	Radial basis function
ROC	Receiver operating characteristic
SD	Standard deviation
SE	Standard error
SES	Socioeconomic status
SHAP	SHapley Additive exPlanations
SVM	Support vector machine
TPR	True-positive rate

Declarations

Availability of data and materials

The PISA microdata analysed here are open data, publicly available from the OECD PISA database (<https://www.oecd.org/pisa/data/>). The raw student- and school-questionnaire files are not redistributed with this work; the full processing, modelling and analysis pipeline that regenerates every processed slice, table and figure is available in the project repository at <https://github.com/xhoja/oecd-pisa-albania>.

Competing interests

The author declares that they have no competing interests.

Funding

Not applicable.

Authors' contributions

X.I. is the sole author and performed all roles: conceptualisation, data curation, methodology, software, formal analysis, interpretation, and writing of the manuscript. The author read and approved the final manuscript.

Acknowledgements

The author thanks Professor Agresa Qosja for her supervision and guidance throughout this project, and the Metropolitan University of Tirana for institutional support. This work draws on the OECD PISA public-use files, and the author thanks the OECD for making these data openly available.

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